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Paper Code : PCC-CS601/PCC-CS603/PCCAI601/PCCCS601 Database Management Systems

UPID : 006577

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) In an ER diagram what is represented by Rectangle?
- (ii) What are the conditions of 3NF?
- (iii) What is metadata?
- (iv) What are the TWO phases in two-phase locking protocol?
- (v) What is MAC model?
- (vi) What are the ACID properties of a transaction?
- (vii) Who is having central control over data and programs accessing that data?
- (viii) What is Relational operator product?
- (ix) Which language is used in application programs to request data from DBMS ?
- (x) What operations can be performed on an item that is having exclusive lock?
- (xi) Which stored procedure is used to test the SQL injection attack?
- (xii) What is the full form of ODBC?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What is 2-phase locking protocol? How does it guarantee serializability? [5]
3. Briefly explain ACID property. [5]
4. Briefly explain the following actions of DBA: [5]
Account creation, Privilege granting, Privilege revocation and Security level assignment.
5. Briefly explain the most significant difference between 2-phase locking protocol and the time-stamping technique concurrency protocol. [5]
6. Explain how the "GROUP BY" clause works. What is the difference between the WHERE and HAVING clauses? Explain them with the help of an example for each. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Consider the following tables:

[8]

RESTAURANTS(r_id, r_name, r_city, phone, seat_capacity)

DISHES(d_id, d_name, d_type, r_id)

CUSTOMER(c_id, c_name, c_city, d_id)

SERVES(r_id, d_id)

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HAS_ALLERGY(c_id, d_id)

i) Find the names of the restaurants that serves "Pizza"

ii) List the names of the dishes that customer "Rony" can eat without allergy problem.

iii) List the names of the dishes that appear in all restaurants in "Delhi"

iv) Find the names of customers who are in the same city as restaurant "Delicious" and who can eat at least one dish at "Delicious" without allergy problem.

(b) Briefly discuss the major problems with performing update operations through "view".

[3]

(c) Specify the following constraints in SQL using accurate applicable command:

[4]

i) All values of attribute "marks" in relation "enrollment" must be less than 200.

ii) If the value of attribute "grade" is 'O' then the mark must be more than or equal to 90. If the value of attribute "grade" is 'E' then the mark must be less than 90 but greater than or equal to 80.

8. (a) Explain various states of the transactions with a state diagram?

[5]

- (b) Briefly explain the requirements of Cascadeless Schedule. [5]
- (c) Discuss the statement- "Transaction cannot be nested inside one another". [5]
9. (a) State two-phase commit protocol and discuss the implications of a failure on the part of: [5]
- The coordinator
 - A participant, during each of the two phases.
- (b) Describe the wait-die and wound-wait protocols for deadlock prevention. [5]
- (c) Let T1, T2 and T3 be transactions that operate on the same data items A, B and C. Let, Ri(A) means that Ti reads A, Wi(A) means that Ti writes A, where i=1 or 2 or 3. Consider the following, [5]
- S1: R2(C). R2(B). W2(B). R3(B). R3(C). R1(A). W1(A). W3(B). W3(C). R2(A). R1(B). W1(B).W2(A).
- Is the schedule serializable?
10. Consider the following relations: [15]
- Employee(empID, firstName, lastName, address, DOB, sex, position, deptNo)
 Department(deptNo, deptName)
 Project(projNo, projName, deptNo)
 WorkOn(empID, projNo, hours_worked)
- Write the SQL statements for the following:
- List the name and addresses of all employees who work in IT department.
 - List the total hours worked by each employee, arranged in order of department name, alphabetically by employee First Name.
 - List the total number of employees of each department for those departments with more than 10 employees.
 - List all the project number, project name and the number of employees who work on that project.
 - List all the employees who have worked more than 20 hours. <https://www.makaut.com>
11. (a) Explain the purpose of checkpoint mechanism. [5]
- (b) What are the roles of Analysis, Redo and Undo phases in the recovery algorithm 'ARIES'. [5]
- (c) Explain the requirement of query optimization. [5]

*** END OF PAPER ***